

Impact of Headphone Quality on Remotely Delivered Repetition Tasks

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Rationale

Speech repetition tasks are a staple of language evaluations in both clinical and research settings. In recent years, there has been increased interest in online testing. In a pilot study, we found that successfully passing a hardware audio check was significantly correlated with performance on nonword repetition tasks (e.g. Dollaghan and Campbell, 1998), but not on real word repetition (e.g. Catts, 1986). An explanation for this finding is that participant’s preexisting lexical knowledge can compensate for potential information loss from low quality hardware. Our research questions were:

- 1. Is performance on nonword repetition negatively impacted by lower headphone quality, but not real word tasks?
- 2. Do the results of this study suggest anything about the equipment required for completing speech repetition tasks for online research studies?

Participants

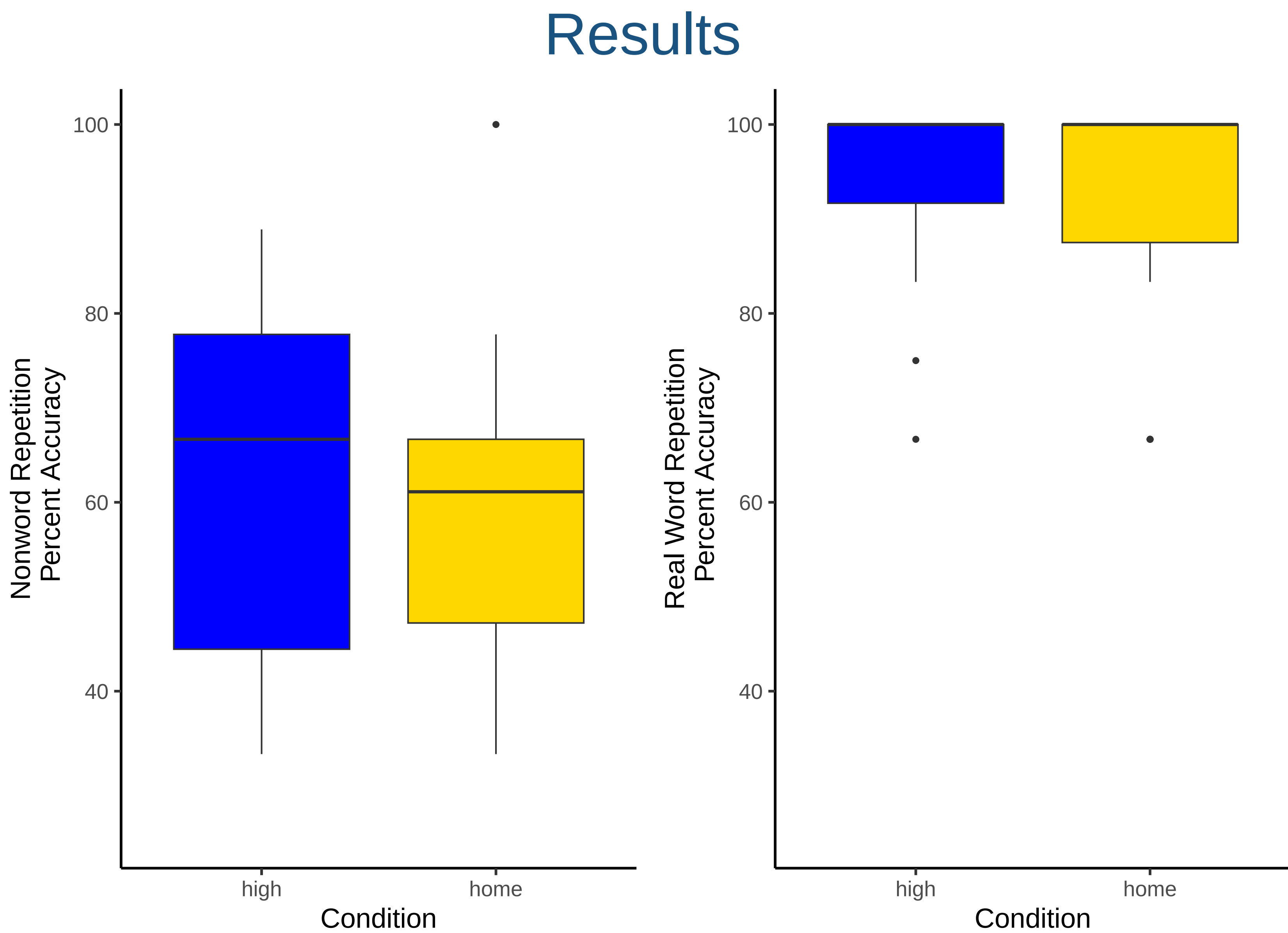
We recruited **20 typically developing adults** (19-64 years old). Our **inclusion criteria** were: (1) Native speaker of an American English dialect, (2) Educated in the United States, and (3) No history of hearing loss, neurological trauma, or cognitive decline.

N = 19
Mean age = 27.35 years (Range = 19-50)
Self-reported Gender: Female = 13; Male = 7
Self-reported Race: White = 15; Other * or >1 = 5

*Due to low counts (less than 4), we are had to collapse Black, Asian, and more than one race.

Task	Low	High
Audio Check	5.37 (1.73)	4.73 (2.18)
Real Word Repetition	90.35 (13.39)	93.86 (9.56)
Nonword Repetition	57.31 (19.69)	56.14 (23.27)

HEADPHONE QUALITY DOES NOT IMPACT PERFORMANCE ON NONWORD OR REAL WORD REPETITION TASKS



Task	Condition	List * Time
Audio Check	t (18) = -1.88 p = .076	---
Real Word Repetition	t (18) = 1.12 p = .279	F (3, 34) = 1.29 p = .295
Nonword Repetition	t (18) = -0.24 p = .810	F (3, 34) = 0.10 p = .286

Procedures and Scoring

Procedures. Tasks were coded using jspsych and delivered remotely on JATOS. Participants completed these tasks in the lab on a laptop computer with two headphone conditions: home quality earbuds (low) and lab quality headphones (high), counterbalanced between participants. During testing, participants performed an audio check with their first hardware condition, then completed half of the word lists for our nonword and real word repetition tasks. The hardware was then switched, and the tasks were repeated.

Task	Measures	Time
Audio Check	Hardware	2 mins
Real Word Repetition	Working Memory	2 mins
Nonword Repetition	Phonological Loop	2 mins

Scoring. Participant performance for each task was scored by an RA. Items were scored as correct/incorrect. Incorrect responses were transcribed. Audio check was automatically scored.

Analysis. To test our hypothesis, we used within subject t-testing to compare performance on the repetition tasks between conditions (low vs high). We used regression models to check for list and time main effects and interactions.

Conclusion

Our results indicate that headphone quality is not a significant barrier in producing usable data for online speech repetition tasks.

In future studies, we will explore **internet speeds** and **individual differences** as alternative explanations for the correlation between audio check and nonword repetition.

References

Catts, H. W. (1986). Speech Production/Phonological Deficits in Reading-Disordered Children. *Journal of Learning Disabilities*, 19(8), 504–508. <https://doi.org/10.1177/002221948601900813>
Dollaghan, C., & Campbell, T. F. (1998). Nonword Repetition and Child Language Impairment. *Journal of Speech, Language, and Hearing Research*, 41(5), 1136–1146. <https://doi.org/10.1044/jslhr.4105.1136>